

Q1

Answers

General comments/theory/notation:

The first thing to note here is that the table contains **par yields**, not yields on zero-coupon bonds. Recall the par yield is the coupon rate for which the price of a bond equals its face value.

Let $P_b(T)$ denote the price in 2023 of a zero-coupon bond with face value \$1 and time to maturity T .

Relationship between par yields and ZCB prices:

$$1 = \sum_{t=1}^T c(T)P_b(t) + 1 \cdot P_b(T)$$
$$\Leftrightarrow P_b(T) = \frac{1}{1 + c(T)} \left(1 - c(T) \sum_{t=1}^{T-1} P_b(t) \right)$$

Continuously compounded yields on zero coupon bonds satisfy $y_b(T) = -\frac{1}{T} \ln(P_b(T))$

The continuously compounded forward rate $y_f(T_1, T_2)$ from T_1 to T_2 is calculated as:

$$\exp((T_2 - T_1)y_f(T_1, T_2)) = \frac{\exp(T_2 y_b(T_2))}{\exp(T_1 y_b(T_1))}$$

[25 points total: 5 points each part]

a) From the relationship between ZCB prices and par yields, we have

$$P_b(1) = \frac{1}{1 + c(1)} = \frac{1}{1 + 0.0487} = 0.95356155239 \approx 0.9536$$
$$P_b(2) = \frac{1 - c(2)P_b(1)}{1 + c(2)} = \frac{1 - 0.0436 \cdot 0.9536}{1 + 0.0436} = 0.91838320842 \approx 0.9184$$

The price of a 2-year zero coupon bond that pays \$100 in 2025 is $100 \times P_b(2) = \$91.84$.

Grading:

- 5pt: correct
- 4pt: correct for face value \$1, but does not say for face value of \$100
- 3pt: correct equations but wrong calculation
- 2pt: tried to specify but some errors in the equation and answer
- 0pt: read the table as ZCB yields

b)

$$P_b(3) = \frac{1 - c(3)(P_b(1) + P_b(2))}{1 + c(3)} = \frac{1 - 0.0412 \cdot (0.9536 + 0.9184)}{1 + 0.0412} = 0.88635792917 \approx 0.8864$$

The price of a 3-year zero coupon bond that pays \$100 in 2026 is $100 \times P_b(3) = \$88.64$.

Grading:

- 5pt: correct OR correct but knock-on calculation error based on previous answer
- 4pt: correct for face value \$1, but does not say for face value of \$100
- 4pt: correct equations but minor calculation error
- 3pt: correct equations but other wrong calculation
- 2pt: tried to specify but some errors in the equation and answer
- 0pt: read the table as ZCB yields

c) The price of an asset that pays \$50 in 2025 and \$200 in 2026 is:

$$50 \times P_b(2) + 200 \times P_b(3) = 50 \times 0.9183 + 200 \times 0.8864 = \$223.19$$

Grading:

- 5pt: correct OR correct but knock-on calculation error based on previous answer
- 4pt: correct equations but minor calculation error
- 3pt: correct equation but wrong calculation OR equation not directly specified but presumed correct in terms of wrongly plugged in ZCB bonds from table
- 2pt: tried to specify but some errors in the equation and answer

d) Recall that the continuous yield for a zero-coupon bond is

$$Ty_b(T) = \ln \left(\frac{1}{P_b(T)} \right)$$

So the continuously compounded forward rate from 2024 to 2026 we are looking for is

$$\begin{aligned} \exp(2y_f(1,3)) &= \frac{\exp(3y_b(3))}{\exp(y_b(1))} = \frac{P_b(1)}{P_b(3)} \\ y_f(1,3) &= \frac{1}{2} \ln \left(\frac{P_b(1)}{P_b(3)} \right) = \frac{1}{2} \ln \left(\frac{0.9536}{0.8864} \right) = 0.03654158285 \approx 3.65\% \end{aligned}$$

Alternatively

$$\begin{aligned} y_b(1) &= \ln \left(\frac{1}{P_b(1)} \right) = \ln \left(\frac{1}{0.9536} \right) = 0.0475513 \approx 0.04755 \\ y_b(3) &= \frac{1}{3} \ln \left(\frac{1}{0.8864} \right) = 0.040196 \\ y_f(1,3) &= \frac{1}{2} (3y_b(3) - y_b(1)) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} (3 \cdot 0.040196 - 0.04755) \\
&= \frac{1}{2} \left(\ln \left(\frac{1}{P_b(3)} \right) - \ln \left(\frac{1}{P_b(1)} \right) \right) = \frac{1}{2} \ln \left(\frac{0.9536}{0.8864} \right) = 0.03654158285 \approx 3.65\%
\end{aligned}$$

Grading:

- 5pt: correct OR correct but knock-on calculation error based on previous answer
 - 4pt: correct equations but minor calculation error
 - 3pt: correct equations but other wrong calculation (e.g. answering correctly for the annually compounded instead of continuous compounded yield)
 - 2pt: tried to specify but some errors in the equation and answer
- e) Flood insurance pays \$100,000 if your house floods in 2024 (probability of flood is 2%), and 0 otherwise. To be priced fairly, the present value of flood insurance must be its price.

$$P(\text{flood}) \times \text{Payout} \times P_b(1) = 2\% \times \$100,000 \times 0.95356155239 \approx \$1907.12$$

Grading:

- 5pt: correct OR correct but knock-on calculation error based on previous answer
- 4pt: correct equations but minor calculation error
- 3pt: correct equations but other wrong calculation
- 3pt: set up the problem using SDF/risky asset terminology including equations and said that more information is needed
- 2pt: tried to specify but some errors in the equation and answer

Q2

Answers

- a) Let's denote the SDF of state s , time t as $M_t(s)$. The price of the 1-year zero coupon bond with the face value of \$1 is $P_b(1) = 1 * M_1(1) * \pi_1(1) + 1 * M_1(2) * \pi_1(2) = 0.58 * 0.8 + 2.6 * 0.2 = 0.98$.

Grading:

- 5pt: correct
- 3pt: correct equation but wrong calculation

- b) The price of the 2-year zero coupon bond with the face value of \$1 is $P_b(2) = 1 * M_2(1) * \pi_1(1) + 1 * M_2(2) * \pi_1(2) = 0.55 * 0.8 + 2.5 * 0.2 = 0.94$. $P_b(2) = \frac{1}{(1+y_b(2))^2}$. $y_b(2) = 3.14\%$

Grading:

- 5pt: correct
- 3pt: correct equations but wrong calculation
- 2pt: correct equation and answer for either the zero coupon bond price or the yield

- c) $P_A = \sum_{t=1}^2 E[M_t X_t] = 110 * 0.58 * 0.8 + 80 * 2.6 * 0.2 + 115 * 0.55 * 0.8 + 85 * 2.5 * 0.2 = 185.74$.

Grading:

- 5pt: correct
- 3pt: correct equations but wrong calculation
- 1pt: Used 4 right pairs of cash flow and state price ($M * \pi$) in the equation for $P(A)$ but one error in the equation (e.g., average instead of sum). No point given in the case of separately miscalculating $P(A)$ in year 1 and year 2

- d) $P_B = \sum_{t=1}^2 E[M_t X_t] = 100 * 0.58 * 0.8 + 90 * 2.6 * 0.2 = 93.2$.

Grading:

- 5pt: correct
- 3pt: correct equations but wrong calculation
- 1pt: Used 4 right pairs of cash flow and state price ($M * \pi$) in the equation for $P(B)$ but one error in the equation (e.g., average instead of sum)

- e) The expected price of B in year 1 is $100 * 0.8 + 90 * 0.2 = 98$. The annually compounded expected return on B is $r = 98/93.2 - 1 = 5.15\%$.

Grading:

- 5pt: correct

- 3pt: correct equations but wrong calculation
- 2pt: Correct equation and answer for either the expected selling price or expected return

Q3

Answers

- a) Both bonds are sold for the same price but bond A pays an extra dollar each period. So, there is **Arbitrage** by shorting one unit of bond B and longing 1 unit of bond A.

Grading:

- 5 points: correct direction and units
 - 2 points: correct direction, but incorrect units
 - 1 point: correct units, but incorrect direction
 - 0 points: incorrect
- b) It can be shown that for par bonds the yield equals to the coupon rate. Hence both C and D are par bonds and trade for 100. You can verify this by calculating their prices using the price-yield equation. Therefore, there is **Arbitrage** as in the first question, short 1 unit of C and longing one unit of D.

Grading:

- 5 points: correct direction and units
 - 2 points: correct direction, but incorrect units
 - 1 point: correct units, but incorrect direction
 - 0 points: incorrect
- c) Again E is a par bond since its yield equals to its coupon rate. Hence, its price is 100. For F since its yield is higher than its coupon rate then its price is lower than 100. But there is **No Arbitrage** opportunity because the principal payment for E comes 5 years earlier than F.

Note: If we assume that the ZCBs also are traded in the background then the answer could be **Unknown**

Grading:

- 5 points: correct answer
 - 0 points: otherwise
- d) Creating a portfolio comprised by 10 units of B and 5 units of C, we can exactly replicate A. But the price of this portfolio is 1 \$. Hence, there is **Arbitrage** by shorting the aforementioned portfolio of B and C, and longing a unit of A.

Grading:

- 5 points: correct direction and units for all assets
- 2 points: incorrect direction but correct units/ incorrect units but correct direction for all three assets

- 1 point: arbitrage only for one state
- 0 points: otherwise

Q4

Answers

a) Step 1: Compute the excess returns for the risky assets.

$$\begin{aligned}\mathbb{E}[R_{USA} - R_b] &= 0.87\% - 0.08\% = 0.79\% \\ \mathbb{E}[R_{Canada} - R_b] &= 0.89\% - 0.08\% = 0.81\%\end{aligned}$$

Step 2: Set up the variance-covariance matrix of the risky assets.

$$\Sigma = \begin{bmatrix} (4.30\%)^2 & 0 \\ 0 & (5.65\%)^2 \end{bmatrix}$$

Step 3: Compute the inverse of the variance-covariance matrix.

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{(4.30\%)^2} & 0 \\ 0 & \frac{1}{(5.65\%)^2} \end{bmatrix}$$

Step 4: Compute the vector of portfolio weights for the tangency portfolio.

$$\begin{aligned}\hat{w} &= \frac{\Sigma^{-1} \mathbb{E}[R - R_b]}{\mathbf{1}^T \Sigma^{-1} \mathbb{E}[R - R_b]} = \frac{\begin{bmatrix} \frac{1}{(4.30\%)^2} & 0 \\ 0 & \frac{1}{(5.65\%)^2} \end{bmatrix} \begin{bmatrix} 0.79\% \\ 0.81\% \end{bmatrix}}{\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{(4.30\%)^2} & 0 \\ 0 & \frac{1}{(5.65\%)^2} \end{bmatrix} \begin{bmatrix} 0.79\% \\ 0.81\% \end{bmatrix}} \\ &= \frac{\begin{bmatrix} 4.273 \\ 2.537 \end{bmatrix}}{6.81} = \begin{bmatrix} 0.6274 \\ 0.3726 \end{bmatrix}\end{aligned}$$

Therefore, the portfolio weights for the tangency portfolio are USA 62.74% , Canada 37.26%, Riskless 0%.

Grading:

- +2 points: Correctly sets up the problem by using the correct equations
 - +1 point: Partially wrote correct equation
- +2 points: Correctly substitutes the numbers into the equations
 - +1 point: Partially substituted numbers correctly
- +2 points: Carries out correct computations to get the right weights
 - +1 point: Correctly computed weights

b)

Step 1: Compute the expected excess return of the tangency portfolio:

$$\mathbb{E}[R_p - R_b] = \hat{w}_{USA} \mathbb{E}[R_{USA} - R_b] + \hat{w}_{Canada} \mathbb{E}[R_{Canada} - R_b]$$

$$\begin{aligned}
&= 0.6274 \cdot 0.79\% + 0.3726 \cdot 0.81\% \\
&= 0.797\%
\end{aligned}$$

Step 2: Compute the standard deviation of the tangency portfolio.

$$\begin{aligned}
\text{Var}(R_p) &= \hat{w}_{USA}^2 \text{Var}(R_{USA}) + \hat{w}_{Canada}^2 \text{Var}(R_{Canada}) + 2\hat{w}_{USA}\hat{w}_{Canada} \text{Cov}(R_{USA}, R_{Canada}) \\
&= 0.6274^2 \cdot (4.30\%)^2 + 0.3726^2 \cdot (5.65\%)^2 + 2 \cdot 0.6274 \cdot 0.3726 \cdot 0 \\
&= 0.001171 \\
\sigma(R_p) &= \sqrt{\text{Var}(R_p)} = \sqrt{0.001765} = 3.42\%
\end{aligned}$$

Step 3: Compute the Sharpe ratio of the tangency portfolio.

$$\begin{aligned}
\text{Sharpe ratio} &= \frac{\mathbb{E}[R_p - R_b]}{\sigma(R_p)} \\
&= \frac{0.797\%}{3.42\%} \\
&= 0.233
\end{aligned}$$

Therefore, the Sharpe ratio for the tangency portfolio is 2.33.

Grading:

- +2.5 points: Correctly computes the expectation of portfolio returns
 - +1 point: Partial credit for expectation calculation
- +2.5 points: Correctly computes the standard deviation of portfolio returns and reports them
 - +1 point: Partial credit for standard deviation calculation
- +1 points: Correctly computes Sharpe ratio

c) For the 2-asset case, we solve the minimization problem as:

$$\min_w w^2 \text{Var}(R_1) + (1 - w)^2 \text{Var}(R_2)$$

We solve for w using the first-order condition:

$$2w \text{Var}(R_1) - 2(1 - w) \text{Var}(R_2) = 0$$

We rearrange and obtain:

$$w = \frac{\text{Var}(R_1)}{\text{Var}(R_1) + \text{Var}(R_2)} = \frac{0.0565^2}{0.043^2 + 0.0565^2}$$

So $w_{USA} = 0.633$, $w_{Canada} = 1 - w_{USA} = 0.367$.

Grading:

- +3 points: Correctly set-up minimum variance problem

- +3 points: Correctly solve minimum variance problem and obtain correct weights
- +1 point: Partial credit given if student incorrectly sets up and attempts to solve minimum variance problem

d) We need to determine which of the two portfolios (A or B) can be an optimal portfolio for an investor, without knowing their risk tolerance. Recall that the optimal portfolio for an investor is a combination of the tangency portfolio and the riskless asset, depending on the investor's risk tolerance. The tangency portfolio weights we calculated in a) were:

USA: 62.74%; Canada: 37.26%; Riskless: 0%

For Portfolio A, the relative weight of USA: $31\% / (31\% + 19\%) = 62\%$, whereas relative weight of Canada: $19\% / (31\% + 19\%) = 38\%$

For Portfolio B: relative weight of USA: $25\% / (25\% + 22\%) = 53.19\%$; Relative weight of Canada: $22\% / (25\% + 22\%) = 46.81\%$

We can see that the relative weights of the risky assets in Portfolio A are much closer to the tangency portfolio weights than those of Portfolio B. Neither are identical.

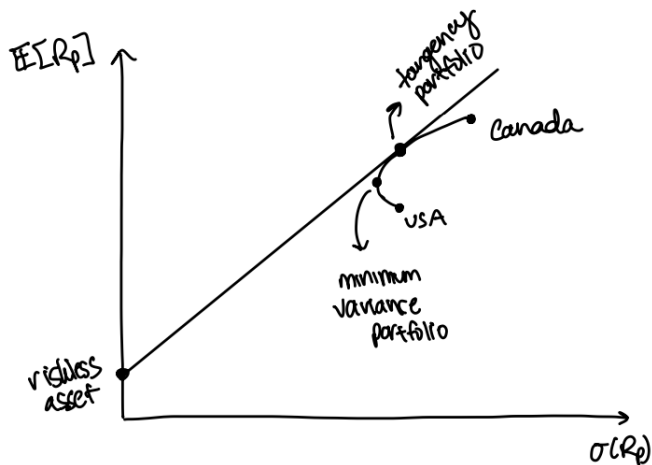
Strictly speaking, if these numbers are correct, then neither could be an optimal portfolio. If we take into account that there might have been rounding errors, portfolio A has relative weights of risky assets that are very close to the tangency portfolio. Therefore, Portfolio A is closer to an optimal portfolio for an investor with some specific risk tolerance.

On the other hand, Portfolio B has relative weights of risky assets that are quite different from the tangency portfolio. This means that Portfolio B cannot be obtained as a combination of the tangency portfolio and the riskless asset. Therefore, Portfolio B cannot be an optimal portfolio for any investor, regardless of their risk tolerance.

Grading:

- +2 points: Carries out thorough computations for portfolio A
- +2 points: Carries out thorough computations for portfolio B
 - +2 points: Partial credit for brief calculations for portfolios A and/or B
- +2 points: Correctly reasons that portfolio A is closer to optimal

e) Below is the diagram with all requested components:



A portfolio that invests only in the USA is not efficient because, due to the curvature of the efficient frontier, it is possible to choose another portfolio that has the same variance and higher expected return. This can be shown by drawing a vertical line through the USA asset point. Since correlation between the two risky assets is zero ($\rho = 0$), the efficient frontier will take on the curved shape shown in the diagram.

Grading:

- + 1 point: correct drawing of riskless asset and tangency portfolio at the efficient frontier curve
- + 1 points: correct designation of the US and CAN assets as endpoints of the frontier
- +1 point: correctly draw min variance portfolio(needs to be distinct of US)
- +3 points: explain inefficiency of the US asset by comparing to other attainable points that strictly dominate it in terms of mean/variance or Sharpe ratio)
- + 1 point: explain inefficiency of investing only on US by appealing to diversification with the uncorrelated CAN asset
- 0 points: incorrect